

Fatemeh Fardno

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Research Interests Reinforcement learning, Game Theory, Optimization, Multi-Agent Systems, Online learning

Education

University of Illinois Urbana-Champaign	Champaign, IL, USA
PhD in Electrical and Computer Engineering	Jan 2024 – Present
Advisor: S. Rasoul Etesami. <i>GPA: 4.0.</i>	

University of Waterloo	Waterloo, ON, Canada
MASc in Electrical and Computer Engineering	Sep 2020 – Sep 2022
Advisor: S. Majid Zahedi. <i>GPA: 3.8.</i>	

Sharif University of Technology	Tehran, Iran
BS in Electrical Engineering	Sep 2015 – Dec 2019
Advisor: Mohammad Saleh Tavazoei. <i>GPA: 3.87.</i>	

Honors and Awards

James W. Beauchamp Award for Interdisciplinary Research	2026
University of Illinois Urbana-Champaign	
Best Poster Award	2026
21st CSL Student Conference	
International Master's Award of Excellence	2020–2022
University of Waterloo	
Membership of National Elite Foundation	2015–2019
Sharif University of Technology	
3rd Prize in the 15th Austrian Physicists' Tournament (AYPT)	2013
1st Prize in the 5th Persian Physicists' Tournament (PYPT)	2012

Ongoing Work

Efficient and Independent Learning of Nash Equilibrium Policies in Markov Aggregative Games
with S. Rasoul Etesami.
Developed a decentralized reinforcement learning algorithm for learning Nash equilibria in stochastic games.

RLHF with General Preference Models and Inference-Time MPC Planning
with Vishesh Prasad, Rohan Deb, and Arindam Banerjee.

Developed a reinforcement learning from human feedback (RLHF) framework that replaces traditional scalar reward models with general pairwise preference predictors, enabling richer preference representations and inference-time planning via model predictive control (MPC).

Publications

Dynamic Transaction Scheduling and Pricing in the Ethereum Mempool

Fatemeh Fardno, S. Rasoul Etesami.

Under Review.

A Game-theoretic Framework for Distributed Load Balancing: Static and Dynamic Game Models

Fatemeh Fardno, S. Rasoul Etesami.

Automatica, Volume 185, March 2026.

Beyond Theorems: A Counterexample to Potential Markov Game Criteria

Fatemeh Fardno, S. Majid Zahedi.

Under Review.

Power allocation of sensor transmission for remote estimation over an unknown Gilbert-Elliott channel

Tahmoores Farjam, Fatemeh Fardno, Themistoklis Charalambous.

2020 European Control Conference (ECC).

Research Experience

Graduate Research Assistant at the Coordinated Science Laboratory (CSL), UIUC

Mentor: Professor S. Rasoul Etesami

Jan 2024 – Present

Graduate Research Assistant at the Multi-agent Systems Lab (MASLAB), University of Waterloo

Mentor: Professor S. Majid Zahedi

Sep 2020 – Sep 2022

Conducting research at the intersection of multi-agent reinforcement learning and game theory, with applications to distributed load balancing.

Thesis title: [Fictitious Mean-field Reinforcement Learning for Distributed Load Balancing](#)

Research Intern at the Distributed Systems Control Group, Aalto University

Mentor: Professor Themistoklis Charalambous

Jun 2019 – Sep 2019

Designed a sensor scheduling policy for remote state estimation over time-varying communication channels. Modeled the system as a partially observable Markov decision process (POMDP) and applied Bayesian inference to learn channel statistics.

	Research Assistant at the Superconductor Electronics Research Laboratory (SERL), Sharif University of Technology Mentor: Professor Mehdi Fardmanesh Sep 2017 – Sep 2018 Fabricated microfluidic chips for high-sensitivity electrical measurements.	
Teaching Experience	Teaching assistant, ECE, UIUC Fall 2024 – Present ECE 110: Introduction to Electronics ECE 313: Probability with Engineering Applications ECE 314: Probability in Engineering Lab	
	Teaching assistant, ECE, University of Waterloo Fall 2021 – Spring 2022 ECE 108: Discrete Mathematics and Logic	
	Teaching assistant, ECE, Sharif University of Technology Fall 2019 Engineering Statistics and Probability	
Industry Experience	Microchip Technology Inc., FPGA Business Unit Toronto, ON, Canada Software Engineer Oct 2022 – Dec 2023 Full-time software engineer in the Timing, Power and Device Data Group. Built and deployed an XGBoost-based delay prediction model for post-synthesis netlists. Improved internal tooling through code reviews, unit tests, and software quality enforcement. Collaborated cross-functionally with hardware and verification teams.	
	Dana Control Tehran, Iran Control Engineering Intern Summer 2018 Programmed PLC systems and designed HMI interfaces for industrial automation.	
Poster Presentations	A Game-Theoretic Framework for Distributed Load Balancing: Static and Dynamic Game Models Midwest Workshop on Control and Game Theory (MWC GT), April 2026 CSL Student Conference, February 2026 Allerton Conference, September 2025	
Skills	Programming Proficient in Python, C/C++, SQL, MATLAB, and Verilog.	
	Languages Farsi, English, Turkish	
Service	UIUC ECE GradSAC Sep 2024 – Present	

Represent graduate student interests in departmental discussions. Support organization of graduate student events and community-building activities.

Reviewer

Fall 2025 – Present

Automatica, IEEE Transactions on Automatic Control (TAC), Journal of Optimization Theory and Applications (JOTA), IEEE Conference on Decision and Control (CDC)